Responsive Redesign

Due Mar 13 and Mar 20 at 11:30pm | Hand in on Canvas

Overview

The goal of this assignment is for you to practice the workflow of redesigning a web page. You should take away the skills necessary to analyze and identify flaws in an existing interface, create mockups for various screen sizes, and build a responsive website based on those prototypes.

Timeline

Task	
Checkpoint Handin: Parts 1 and 2	Thu 3/13 11:30pm
Critique II	Tue 3/18 or Wed 3/19
Final Handin: All work	Thu 3/20 11:30pm

Note that this is a **heavy** assignment. In prior years, students have noted that this assignment (especially Part 3) is not something they were able to complete in a week, and recommended that future students be told this. We recommend sticking to the above with a "checkpoint" to ensure that you can complete the assignment on time. This **checkpoint is worth 1 point** that is designed to be a minimum milestone to help you pace.

Part 1: Identifying Usability Problems

Picking a Webpage

Pick **one page** of a website where the link is viewable by anyone without creating an account. Avoid Brown-specific sites like Banner, Workday, and ASK, where your perspective as one of the stakeholders may be skewed and would be a poor portfolio piece because people outside Brown can't access or understand the original design. We recommend avoiding mass consumer websites with dedicated UI/UX designer(s), like Facebook or Twitter, as you will have a harder time convincingly improving the design of those pages. Instead, choose a less popular interface to improve, like websites for a local hospital, college application or fellowship websites, network router support pages, a custom-made restaurant ordering site, local government service, etc.

You will have an easier time with the rest of the assignment and be able make more substantial improvements if you pick a simple yet poorly-designed website to start. Picking a simpler page will make redesigning it a lot easier later, so keep that in mind. Feel free to check your chosen interface with a TA. **After you choose an interface to redesign, take a screenshot and link to it. Write a sentence about why you chose this website.**

Finding Problems

Analyze the usability of the web page using the criteria we learned about in class: efficiency, learnability, and memorability, and think about its conceptual model. **Write about these problems**, in the form of a list, table, infographic, or bullet points (depending on how you would like to present it in your portfolio)**.** It might be helpful to refer back to some of the earlier classes.

In addition to analyzing the website on the three usability guidelines, use WebAIM WAVE to detect possible accessibility problems. Do you agree with the problems detected? **Write a few sentences about your findings.**

Part 2: Visual Redesign

Based on the problems you identified in Part 1 and the page that you chose, create **a visual design style guide, and mockups**. **Convey through annotations or explanations how you address the usability and accessibility issues you identified in Part 1**.

Visual Design Style Guide

Clear documentation makes your design work more consistent and makes handoffs to other members on a team easier. **Before you start your mockups, create a visual design style guide (e.g., in Figma) that displays the main colors, typography, and reusable components' different states (buttons on hover, click, etc.)**. Here is an example.

Major components should be included in your style guide, but how many components this totals to will vary based on your design choices. For interactive elements (e.g., elements that change on hover, on click), keep their interactions consistent throughout your product to keep the user experience coherent. You may find using Figma components helpful for reusing components in your mockup. For more information, refer to the documentation.

Mockups

Using Figma or other mockup tool (like Adobe XD), create a total of 3 mockups of your screen, one for each of the screen sizes (mobile, tablet, desktop). Test on a phone (~375x667px), a tablet (~768x1024px) and a large computer (~3840x2160px), but your interface should work on any dimensions within that range! You are encouraged to sketch out how you may lay out the components before making the mockups; this will actually save you time from having to change your mockups. Design a website that is actually responsive: reusing most of it between screen sizes, rather than designing three fairly different interfaces for different screen sizes.

For each mockup, **annotate the different elements** so that a developer would be able to take your designs and reproduce them without needing additional explanation. Note elements used and key layout choices that are not the default (e.g., whether something is flex, or alignment attributes, or non-obvious margin/padding, or setting width/max-width). **Annotate how your website would be responsive to handle the different screen sizes.** Is it clear how the styling and/or structuring of each element will change when you move from desktop to mobile to tablet? Here are some example questions to consider annotating:

- For larger containers, will you be using flexbox, grid, or something else? Why?
- When in-between sizes, will anything like fonts or spacing need to be adjusted slightly?
- Does the page change because of a user-action (such as a click or a hover)? How?

You've hit a checkpoint! Submit Parts 1 and 2 on Canvas as a PDF by Mar 13 11:30pm as a 1-point completion milestone.

Part 3: Responsive Redesign

Based on the mockups, create the page using HTML and CSS! The goal of this assignment is to get some practice doing this by hand, so CSS libraries are okay (like Bootstrap, Tailwind). Use of AI tools or comprehensive frameworks (like React, Next.js, Vue, Svelte) are not permitted.

The focus of this part is the responsiveness of the redesign. Use a combination of flexbox (or grid) and normal flow to achieve the responsiveness. You are welcome to change text or images from the original web page that you are redesigning. Buttons or navigational elements can be replaced with dummy targets. If there are dropdown menus or other more dynamic elements, you can try to recreate them at least visually, but they do not need to actually be interactive.

If the page is a long scrolling page with substantial redundant parts, you may simplify it for your redesign, but otherwise your redesign should incorporate the full page. While you are welcome to enhance your page with JavaScript or animated/dynamic content, anything overly complex is discouraged if it risks failing to load or becomes brittle during grading on another person's setup.

Test for responsiveness and accessibility:

- Play with the size of your browser window to see how the website looks on other computer screens
- Try changing the font size on your browser, to check that your interface is not too dependent on exact font sizing. Or use Google Translate to see how your website looks in different languages
- Use Developer Tools, e.g. the Chrome Responsive Tool or Firefox Responsive Design Mode to see how the website looks on other devices, like an iPad
- Use WebAIM WAVE to detect accessibility issues

Bring a draft of your design to Critique II, especially if you did not present during Critique I

Once you're ready to submit the project, **deploy your redesigned webpage** and **link** to your responsively redesigned page **in your handin page**. In other words, you will have a page showing your process and narrative (handin page), linking to both the original (one you didn't make) and redesigned page (one you did make). You're welcome to reuse or iterate on your submission page style from a previous assignment to save time. Add any headings or text to guide the reader through your design process. For example, you may want to include screenshots of the redesign in your submission page. There's no need to link your GitHub repo, as your code can be reviewed by the course staff using "View Source" or in Developer Tools.

Final Handin

As usual, you will create a Web page in HTML/CSS that presents your work in a way that stands alone—meaning it should be engaging and informative to anyone interested in design, regardless of whether they know this was a class assignment. Your webpage should provide all necessary context, including what you did, why you did it, and your overall goals, so that viewers can fully understand and evaluate your work on its own merits. Feel free to review the TAs' refresher on Git and HTML/CSS.

The webpage should clearly and concisely communicate the purpose, context, and final product of your project. Make sure the page is visually appealing and easy to navigate, as it should be designed for an audience interested in how you design. Check out our portfolio guide for some examples and guidance on how to structure a portfolio piece. Your handin page will not be evaluated on its responsiveness; only the redesigned page needs to be responsive.

This part can take a lot of time, so think about it early! Once you're finished with creating your webpage, **deploy your webpage** on Github Pages (see the deployment guide), or any host you prefer, like Brown CS, Cloudflare, or Vercel.

Submit both your handin page AND your redesigned page in a single PDF (using Print to PDF). Also **include a link to the redesigned page, and the handin Web page in a Comment on your submission in Canvas** (you can do this after submitting the PDF). Please check that the links work at least until they are graded!

This final handin will be evaluated, whereas the March 13 checkpoint is just checked for completion. In grading the final handin, the course staff looks at both the PDFs and the deployed version for grading. If you're having issues with the PDF layout, try "changing the scaling of the print render in the printer settings", "setting the margin to none", or "setting the paper type to A2 in the Print Dialog". Feel free to ask questions on Ed or visit TA office hours as well.

Rubric

Component	Points
Part 1: Identifying Usability Problems ☐ Organization effectively communicates pre-existing interface issues to any reader? ☐ Understands how usability and accessibility issues of the pre-existing interface impact the user experience?	2
Part 2: Visual Redesign ☐ Visual Design Style Guide is standalone such that no additional context or annotations are needed to apply the style guide to a real component? ☐ Understands how to design consistent visual cues that keep the user experience coherent across components and screens? ☐ Understands how to design intuitive layouts using reusable components, while ensuring smooth transitions across three different screen sizes? ☐ Understands how to effectively communicate the different elements needed to smoothly transition from one screen size to another? ☐ Understands how different user actions impact the page and how to accommodate for various user actions across screens? ☐ All three mock-ups are reproducible with no additional explanation needed?	6
Checkpoint Handin: Parts 1 & 2 (Due March 13th, 11:30 PM)	1
Part 3: Responsive Redesign ☐ Redesigned page works visually as expected on and in between various screen sizes? ☐ Redesigned page works visually as expected on and in between various font sizes or languages for the three standard formats? ☐ Clearly conveys how redesign impacts users and addresses usability issues in Part 1? ☐ Clearly conveys how redesign impacts users and addresses accessibility issues in Part 1? ☐ Redesigned page reflects the high-fidelity mockups and is usable in production?	6
Handin Web Page Portfolio Readiness ☐ Provide enough **context** so that someone who is not taking this course could understand the project. The portfolio item should stand alone without referencing the assignment. ☐ Has an **engaging narrative**, e.g. relates to the reader, interesting lessons learned, a surprising finding, etc. ☐ Has a **clear hierarchy**, e.g. content is laid out in a reasonable order to read, similar items have a similar look, headings are clearly different from content, etc. ☐ Is **concise**, avoids large blocks of text, preferring bullet points and captions. ☐ All included links, images, and PDFs have the appropriate access permissions	2
Critique II	1
Total	**18**